

Table D.3. Expanded instance-segmentation image inventory

Split	Images	Convention-matched images	Added unmatched-name images
Train	1,147	905	242
Validation	144	115	29
Test	161	129	32
Total	1,452	1,149	303

The 303 added images do not expose compatible specimen identity in their filenames. They were therefore allocated by image-level stratified random splitting before the partitions were merged. The expanded 1,452-image inventory must not be described as fully specimen-leakage-free: specimen-group separation is enforced for convention-matched images, while specimen identity cannot be enforced for the unmatched-name subset.

Group-overlap definition and leakage study

For a split strategy, train-test overlap is measured at specimen-group level as

$$\text{GroupOverlap}(\%) = 100 \times \frac{|G_{\text{train}} \cap G_{\text{test}}|}{|G_{\text{test}}|}. \quad (\text{D.1})$$

This quantity is not image overlap. The retained multi-seed comparison uses seeds 42, 123, and 3407 and reports mean $\pm$ standard deviation.

Table D.4. Leakage comparison across retained segmentation split policies

Split policy	Train-test overlap (%)	Full mAP50	Labeled mAP50	mAP50-95	Healthy FP (%)
Plain random image	98.7 $\pm$ 1.2	67.1 $\pm$ 1.3	67.7 $\pm$ 1.3	29.9 $\pm$ 2.4	5.4 $\pm$ 2.7
Stratified random image	94.0 $\pm$ 6.1	65.1 $\pm$ 6.0	65.9 $\pm$ 6.2	28.2 $\pm$ 1.6	9.8 $\pm$ 2.4
Stratified grouped specimen	0.0 $\pm$ 0.0	45.1 $\pm$ 2.7	47.3 $\pm$ 3.5	14.4 $\pm$ 2.1	23.6 $\pm$ 11.0

The higher random-split scores coincide with substantial specimen overlap and therefore cannot be interpreted as stronger leakage-safe generalization. The grouped split is the principal validity reference for the filename-compatible subset.

Appendix E - Attention Configuration Ledger

This appendix preserves both the protocol ledger and the retained parameter/timing evidence for segmentation attention configurations. The broader attention-family comparison is configuration-level evidence, not a strict one-factor ablation, because augmentation, training budget, early stopping, initialization, software versions, placement, and auxiliary objectives are not identical across all rows.

Table E.1. Recorded attention configuration protocols

Config.	Placement / objective	Aug. policy	Epochs (cfg./run)	Initialization	Versions
YOLO11n-seg reference	No added attention	Clean-light	100 / 60, early stop	yolo11n-seg.pt	8.4.61 / 2.10.0
CA→SimAM	P3/P4/P5	Strong	100 / 100	594/594 tensors	8.4.67 / 2.11.0
LKA→SimAM	P3/P4/P5	Clean-light	100 / 75, early stop	579/579 tensors	8.4.61 / 2.10.0
DPCA	P3/P4/P5	Strong	200 / 200	yolo11n-seg.pt	8.4.102 / 2.10.0
CoTEGate + BoundaryLite	P4 + auxiliary loss	Strong	200 / 184, early stop	yolo11n-seg.pt	8.4.95 / 2.10.0

Table E.2. Attention localization-efficiency ledger

Configuration	Parameters	Params (M)	Notebook ms/image	FPS
YOLO11n-seg reference	2,842,998	2.843	6.0	166.7
CA→SimAM	2,854,718	2.855	9.3	107.5
LKA→SimAM	2,963,510	2.964	8.6	116.3
DPCA	2,848,377	2.848	8.8	113.6
CoTEGate + BoundaryLite	2,736,562	2.737	8.2	122.0

These timing values are notebook/GPU diagnostics and are not Android latency. CA→SimAM adds only 11,720 parameters relative to the reference while providing the retained final attention refinement.

Table E.3. CA→SimAM placement study on YOLO11n-seg

Placement	Full mAP50	Lab. mAP50	Lab. mAP50-95	Recall	Params (M)	ms/image	FPS
P3/P4/P5	49.71	55.07	19.64	49.7	2.855	9.3	107.5
Neck concat	40.90	43.30	14.20	46.5	2.819	10.3	97.1
Head P3	40.90	43.40	14.60	41.5	2.692	9.9	101.0
Head P4	47.70	50.50	16.80	47.4	2.693	9.6	104.2
After C2PSA	43.80	45.80	16.90	45.6	2.697	10.3	97.1
Backbone P3	38.50	40.90	13.60	43.6	2.692	10.6	94.3
Neck P3/P4	44.30	47.30	15.90	43.6	2.721	10.4	96.2